

RAMESH PRASAD, Ph.D.

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PROFESSIONAL SUMMARY

Computational biophysicist and AI-driven researcher with a Ph.D. in computational biology and biophysics and **10+ years** of interdisciplinary research experience at the intersection of **protein structure, function, and machine learning**. Develops predictive models using **AlphaFold and XPLOR-NIH** of **protein sequence–structure–function relationships**, biomolecular interactions, and conformational dynamics by integrating **MD simulations, representation learning, and experimental datasets** (NMR, cryo-EM, biophysics). Creator of open computational tools (**ReSMAP, SpiDec**) for protein property prediction and phase-behavior modeling. Applies **deep learning, supervised/unsupervised learning, and statistical modeling** (CNNs, random forests, PCA, JAX) to translate structural and simulation outputs into testable biological hypotheses. Expert in **protein dynamics, membrane proteins, intrinsically disordered proteins (IDPs), biomolecular condensates, and protein–protein/protein–lipid interactions**. Builds scalable **Python/JAX pipelines** on GPU HPC clusters for high-throughput analysis and model development. Author of **28 peer-reviewed publications** including *Nature Communications*, *JACS*, and *Chemical Reviews*. Experienced collaborating with experimentalists, engineers, and computational scientists to advance structure-function discovery.

CORE COMPETENCIES

- **AI & Machine Learning for Protein Science:** Deep learning (CNNs), random forests, PCA, cluster analysis, representation learning from MD trajectories; predictive modeling of sequence–function and structure–function relationships; JAX, scikit-learn, NumPy, SciPy, Pandas
- **Protein Structure, Function & Interactions:** Atomistic MD (AMBER, NAMD, GROMACS), enhanced sampling, protein dynamics, membrane proteins, IDPs, condensates, protein–protein and protein–lipid interactions, homology/integrative modeling (MODELLER, XPLOR-NIH)
- **Sequence–Structure Learning & Predictive Modeling:** Residue-level property prediction; conformational ensemble analysis; integration of AlphaFold/PDB structural models with simulation and experimental restraints; biomolecular interaction energetics (FEP, umbrella sampling)
- **Computational Tool & Pipeline Development:** Open web servers and analysis frameworks (ReSMAP, SpiDec); reproducible Python workflows; Git; Conda; automated trajectory analysis (MDTraj, MDAnalysis)
- **Experimental Data Integration:** Solid-state NMR (PISEMA, DARR), solution NMR (NOE), cryo-EM-assisted modeling; PDB deposition (8U1T, 9NM2); hypothesis-driven validation of computational predictions
- **High-Performance Computing:** Linux HPC (SLURM/PBS), GPU optimization, large-scale simulation campaigns, cloud computing, statistical analysis and visualization

EDUCATION & RESEARCH EXPERIENCE

Visiting Research Specialist, Department of Chemistry, University of Illinois Chicago *Mar 2024 – Present*
Postdoctoral Research Associate, Department of Chemistry, University of Illinois Chicago *Feb 2019 – Jan 2024*

Ph.D., Computational Biology & Biophysics, IACS Kolkata, India *Aug 2012 – Dec 2018*
Thesis: Structure-based inhibitor design and molecular mechanisms of coagulation complexes

Post-M.Sc. Diploma, Bioinformatics, University of Calcutta *2011 – 2012*
M.Sc., Biochemistry, University of Burdwan *2009 – 2011*

SELECTED RESEARCH EXPERIENCE

AI-Driven Protein Sequence–Function & Structure–Property Modeling

- Co-developed **ReSMAP**, a machine-learning-enabled web server predicting **residue-specific membrane-association propensities** of intrinsically disordered proteins from sequence and biophysical features; published in *Membranes* and highlighted in *Annual Review of Biophysics* (2025).
- Built automated Python pipelines for **representation learning from MD trajectories**: contact mapping, hydrogen-bond networks, conformational clustering, and PCA to uncover sequence determinants of protein–lipid binding and IDP function, used AlphaFold as part of integrative structure workflows for membrane-associated proteins (*PLoS Pathogens* 2026; *Biophysical Journal*).
- Applied **supervised ML** (random forests, CNNs) and descriptor-based modeling (RDKit, QSAR) to predict binding affinity, biological activity, and structure–activity relationships in protein–ligand systems.
- Modeled **protein sequence–function relationships** in condensate-forming peptides: linked amino acid composition to phase equilibrium, material properties, and intermolecular interaction networks using MD, QM/DFT, and NMR

(*Cell Reports Physical Science, JACS*).

Protein Structure, Dynamics & Biomolecular Interaction Prediction

- Determined membrane protein structures by integrating **computational structure prediction**, **MD refinement**, and **ssNMR restraints**; deposited SARS-CoV-2 E protein (PDB 8U1T) and *M. tuberculosis* CrgA (PDB 9NM2) (*JACS, Communications Biology*).
- Performed voltage-dependent MD simulations to predict **protein dynamics and ion-transport mechanisms** in NMDA receptors; identified subunit-specific gating behavior and translated findings into mechanistic hypotheses (*Nature Communications*).
- Quantified protein–membrane and protein–protein interaction energetics using umbrella sampling and FEP; investigated lipid-mediated allosteric regulation in coagulation and receptor systems (*ACS Med. Chem. Lett., Integrative Biology*).
- Characterized adhesive peptide Bap1 sequence determinants of membrane binding and lipid-composition-dependent remodeling using enhanced sampling and automated analysis workflows.

Open Computational Tools, HPC Pipelines & Collaborative Research

- Developed **SpiDec**, an open computational framework for predicting **phase boundaries and interfacial tension** of biomolecular condensates from spinodal decomposition simulations (*Frontiers in Molecular Biosciences*).
- Designed scalable GPU/HPC workflows (SLURM/PBS, Conda, Git) for high-throughput MD setup, trajectory analysis, and ML-ready feature extraction across multi-protein campaigns.
- Led interdisciplinary projects integrating **computational predictions with experimental validation** (NMR, cell-based assays); presented at Biophysical Society (2025, 2026) and EMBO IDP Conference (2019); mentored trainees on simulation and Python analysis workflows.
- Peer reviewer and guest editor for protein dynamics, membrane interactions, and drug discovery special issues; contributed reproducible code and datasets supporting published research.

SELECTED PUBLICATIONS

28 total peer-reviewed publications and 2 book chapters. Full list available on [Google Scholar](#).

1. **Prasad***, R., Huang*, X., et al. Conformations and sequence determinants in the lipid binding of an adhesive peptide derived from *Vibrio cholerae* biofilms. **PLoS Pathog** 22, e1013990 (2026). (* joint author) [link](#)
2. MacAinsh, M., Muhammedkutty, F.N.K., **Prasad, R.** and Zhou, H.-X. Membrane association of intrinsically disordered proteins. **Annu. Rev. Biophys.** 54, 275-302 (2025). [link](#)
3. Shin, Y., **Prasad, R.**, Das, N., Taylor, J.A., Qin, H., Hu, M.,..., Zhou, H.-X., and Cross, T.A. Mycobacterium tuberculosis CrgA Forms a Dimeric Structure with Its Transmembrane Domain Sandwiched between Cytoplasmic and Periplasmic β -Sheets, Enabling Multiple Interactions with Other Divisome Proteins. **J Am Chem Soc** 147, 11117–11131 (2025). [link](#)
4. Zhang, Y., **Prasad, R.**, Su, S., Lee, D., and Zhou, H.-X. Amino acid-dependent phase equilibrium and material properties of tetrapeptide condensates. **Cell Rep. Phys. Sci.** 5, 102218 (2024). [link](#)
5. Zhou, H.-X., Kota, D., Qin, S., **Prasad, R.** Fundamental aspects of phase-separated biomolecular condensates. **Chem. Rev.** 124, 8550 (2024). [link](#)
6. Kota, D., **Prasad, R.** & Zhou, H. X. Adenosine Triphosphate Mediates Phase Separation of Disordered Basic Proteins by Bridging Intermolecular Interaction Networks. **J Am Chem Soc** 146, 1326–1336 (2024). [link](#)
7. Kunnath Muhammedkutty, F.N., **Prasad, R.**, Gao, Y., ... Rosenberry, T.L., Paravastu, A.K., and Zhou, H.-X. A common pathway for detergent-assisted oligomerization of A β 42 **Commun. Biol.** 6, 1184 (2023). [link](#)
8. Zhang, R., Qin, H., **Prasad, R.**, Zhou, H.-X. and Cross T.A. Dimeric transmembrane structure of the SARS-CoV-2 E protein. **Commun. Biol.** 6, 1109 (2023). [link](#)
9. Amin, J.B., He, M., **Prasad, R.**, Leng, X., Zhou, H.-X and Wollmuth, L.P. Two gates regulate the kinetics of NMDA receptor activity: Subunit specific regulation of two gates in NMDA receptors **Nature Comm.** 14, 1623 (2023). [link](#)
10. Mazarakos, K., **Prasad, R.**, and Zhou, H.-X. SpiDec: Computing binodals and interfacial tension of biomolecular condensates from simulations of spinodal decomposition. **Front. Mol. Biosci.** 9, 1021939 (2022). [link](#)
11. Qin, S., Hicks, A., Dey, S., **Prasad, R.**, Zhou, H.-X. ReSMAP: Web server for predicting residue-specific membrane-association propensities of intrinsically disordered proteins. **Membranes** 12, 773 (2022). [link](#)
12. Das, K., **Prasad, R.**, Ansari, S.A., Roy, A., Mukherjee, A. Sen, P. Matrix metalloproteinase-2:A key regulator in coagulation proteases mediated human breast cancer progression through autocrine signaling. **Biomedicine & Pharmacotherapy** 105, 395-406 (2018). [link](#)

TECHNICAL SKILLS

AI/ML & Programming: Python, JAX, scikit-learn, NumPy, SciPy, Pandas, Matplotlib, Seaborn; deep learning (CNNs), random forests, PCA, cluster analysis, representation learning, statistical analysis, data visualization

Protein Modeling & Simulation: AMBER, NAMD, GROMACS, XPLOR-NIH, Gaussian (QM/DFT), MODELLER, MDAnalysis, MDTraj, CPPTRAJ

Structure & Interaction Analysis: Integrative/hybrid modeling with PDB and predicted structures; protein dynamics; docking (Glide, AutoDock Vina); RDKit, OpenBabel

Infrastructure: Git, Conda, Linux HPC (SLURM/PBS), GPU clusters, Google Cloud, CMake, Bash, Tcl/Tk

Visualization: PyMOL, VMD, ChimeraX, GaussView

HONORS, PRESENTATIONS & SERVICE

- Best Poster Award, UIC ACER Computational Research Symposium (2026); Small Scholarship Award, UIC (2024)
- Travel Award, EMBO Intrinsically Disordered Proteins Conference (2019); ISTH Reach the World Grant, Dublin (2018)
- DST-INSPIRE Ph.D. Fellowship, Government of India (2012–2017)
- Invited Speaker: Protein membrane mechanisms and vesicle fusion, University of Burdwan (2021)
- Peer reviewer: *Scientific Reports*, *JCIM*, *PROTEINS*, *J. Biomolecular Structure & Dynamics*, and 10+ journals
- Guest Editor, *Biomedicines* special issues on protein dynamics, membrane interactions, and drug discovery (2026)